

2023-Ph-H1-Q3

Section: Our Dynamic Universe

Topic: Energy and Power

Which quantity is conserved in an inelastic collision?

Answer: B

(Momentum)

Guidance for Students:

In **all collisions**, momentum is conserved if there are no external forces. Inelastic collisions involve some loss of kinetic energy (e.g. into heat or deformation).

Revision Tips:

- Momentum is **always** conserved in collisions.
- KE is **only** conserved in **elastic** collisions.
- $p = mv$ applies to each object.
- Total momentum before = total momentum after.